

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Constant Acceleration in 2D

Two-dimensional motion is two one-dimensional motions sharing the same time.

CONSTANT ACCELERATION IN 2D · M1 · 05

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DR ESLAM AHMED · CAIRO UNIVERSITY FACULTY OF ENGINEERING

The Map

This strategy booklet was mined from the local Mechanics 1 topic problem/answer PDFs, Dr Eslam's source notes, and the WME01 website answer bank. Mechanics questions become reliable when the diagram, sign convention, and equation family are chosen first.

01 SPLIT: Split Axes

TRIGGER *2D constant acceleration*

02 PROJ: Projectile Equation

TRIGGER *projectile, trajectory*

03 TIME: Flight Time

TRIGGER *hits ground, returns to level*

04 REL: Relative Position

TRIGGER *two moving particles in 2D*

BANK EVIDENCE Local pages: 19 problem, 39 answer, 4 notes. Website primary entries:
1. Website linked entries: 22.

01 SPLIT: Split Axes

TRIGGER *2D constant acceleration*

BECOMES Resolve into independent i and j equations.

FIRST LINE TO WRITE

$$\mathbf{r} = \mathbf{r}_0 + \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$$

SIMPLEST STRATEGY

1. Separate i and j .
2. Place the same t in both.
3. List known components.
4. Insert acceleration.
5. Tie answers back together.

WORKED MODEL

$$\mathbf{r} = (2 + 3t)\mathbf{i} + (5 - 4.9t^2)\mathbf{j}.$$

HARD VARIANTS

1. Horizontal acceleration is zero in projectile motion unless stated.
2. Use vector components for velocity too.
3. Keep units consistent in both axes.

— BOTTOM LINE

The same time connects both axes.

02 PROJ: Projectile Equation

TRIGGER *projectile, trajectory*

BECOMES Use horizontal motion to express t , then substitute into vertical motion.

FIRST LINE TO WRITE

$$x = u_x t, \quad y = u_y t - \frac{1}{2} g t^2$$

SIMPLEST STRATEGY

1. Pick horizontal equation.
2. Rearrange for t .
3. Output vertical equation.
4. Join them by substitution.

WORKED MODEL

■ If $x = ut$, then $t = x/u$ and $y = u_y(x/u) - \frac{1}{2}g(x/u)^2$.

HARD VARIANTS

1. If launch point is not origin, include initial x, y .
2. Horizontal velocity stays constant only when no horizontal acceleration.
3. Domain matters: the path equation may extend beyond the physical flight.

— BOTTOM LINE

Trajectory comes from eliminating time.

COMMON TRAP Trying to use one SUVAT equation for both axes.

03 TIME: Flight Time

TRIGGER *hits ground, returns to level*

BECOMES Use the vertical displacement condition to find time.

FIRST LINE TO WRITE

$$y(t) = 0 \quad \text{or given height}$$

SIMPLEST STRATEGY

1. Take the vertical equation.
2. Insert final height.
3. Make a quadratic if needed.
4. Exclude the wrong root.

WORKED MODEL

Returning to launch level gives $t = 0$ or $t = \frac{2u_y}{g}$.

HARD VARIANTS

1. If landing height differs, solve the full quadratic.
2. Check that a root lies after launch.
3. Use the time in horizontal motion to get range.

— BOTTOM LINE

The useful time is usually the non-zero positive root.

COMMON TRAP Keeping $t = 0$ as the impact time.

04 REL: Relative Position

TRIGGER *two moving particles in 2D*

BECOMES Subtract position vectors and minimise or solve components.

FIRST LINE TO WRITE

$$\vec{AB} = \mathbf{r}_B - \mathbf{r}_A$$

SIMPLEST STRATEGY

1. Read both position vectors.
2. Extract separation.
3. Link condition: meet, distance, due direction.

WORKED MODEL

■ Due east means the j -component of separation is zero.

HARD VARIANTS

1. Meet means both separation components are zero.
2. Closest distance uses separation squared.
3. Bearings from one object use the relative vector from that object.

— BOTTOM LINE

Relative position turns two particles into one vector.

COMMON TRAP Using absolute positions when a relative vector is needed.

Quick Recognition Table

WHEN YOU SEE	FIRST LINE TO WRITE
2D constant acceleration	$\mathbf{r} = \mathbf{r}_0 + \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$
projectile, trajectory	$x = u_x t, \quad y = u_y t - \frac{1}{2}gt^2$
hits ground, returns to level	$y(t) = 0$ or given height
two moving particles in 2D	$\overrightarrow{AB} = \mathbf{r}_B - \mathbf{r}_A$

Hard Variant Checklist

STRATEGY	CHECK BEFORE FINAL ANSWER
SPLIT: Split Axes	Horizontal acceleration is zero in projectile motion unless stated. Use vector components for velocity too. Keep units consistent in both axes.
PROJ: Projectile Equation	If launch point is not origin, include initial x, y . Horizontal velocity stays constant only when no horizontal acceleration. Domain matters: the path equation may extend beyond the physical flight.
TIME: Flight Time	If landing height differs, solve the full quadratic. Check that a root lies after launch. Use the time in horizontal motion to get range.
REL: Relative Position	Meet means both separation components are zero. Closest distance uses separation squared. Bearings from one object use the relative vector from that object.